

Electronic commerce in the real world

P A Jenkins

There are many challenges facing electronic commerce in the real world. The technical ability to trade electronically may seem to be enough of a challenge on its own to warrant success. However, other issues, like the confidence of users that the technology meets their needs, must also be addressed. This confidence will stem from a number of factors like the interoperability of different implementations, the confidence in the companies providing the infrastructure, and the legal recognition of the resulting agreements to trade. These issues are being addressed in organisations whose activities are aimed at agreeing technical standards, regulations, and business codes of practice. This paper looks at the activities of such organisations and what impact they will have on BT's work.

1. Introduction

Electronic commerce (eCommerce) is a global activity, and while some areas involve two-way communications, in most cases the parties that will need to communicate cannot be predicted in advance. As a result, the way these communications are supported by the technology needs to be agreed internationally. Traditional communications reached similar agreements through formal standardisation bodies; however, the pace at which many of these formal organisations operate in reaching the required agreements is too slow for a fast-moving area like eCommerce. This problem is well recognised, and there are many organisations stepping forward to fill this role. This paper will consider the criteria for choosing a particular body as the avenue to reach such an agreement, and give an outline of the resulting complex relationships.

Addressing the technical issues is only part of the problem. The technical infrastructure used to support open electronic commerce (the Internet) does not reflect administrative and legislative boundaries. The resulting transactions are therefore moving outside the well-understood legal framework within which trading has historically taken place. The issue is not the absence of legislation, but rather the confusion over which of the possible legal systems will apply. Here, too, there is a need for agreement on the applicable framework, but the parties that need to agree include agencies that protect individual rights, and governments, as well as the service provision industry. The different organisations that will be addressing these issues are described, and the use of governmental tools, like regulation, are also considered.

2. What are the external issues?

In an abstract world, a developer who is faced with an abstract task can analyse the problems to be addressed and determine the best solution. This solution can then be implemented in order to complete the task. However, we do not exist in an abstract world and a number of factors will mean that this approach is not appropriate. Some of these factors will be inherent in the task, like cost limitations and time-scales. The developer imposes other constraints, such as the availability of resources to undertake the task. There are also a set of constraints that are determined externally to the developer but will affect the ways in which solutions are produced. It is this group of influences that will be addressed here.

These issues are considered together because the nature of the constraint is determined outside the developer. However, the detail of the constraint imposed, the way in which it affects the design process, and the extent to which a developer such as BT can influence them varies greatly. The following is an attempt to classify these types of issue and the way that they can constrain BT's developments.

- Legislation

Any product or service must accommodate the 'general' legal requirements that are relevant to it. These requirements can be many and varied, including such things as contract law and consumer protection frameworks.

- Regulation

In certain industry sectors there are specific legal requirements imposed on companies providing products or services. Such requirements are termed 'regulations' and any product or service would have to comply with the appropriate requirements, as would the organisation in the way it runs its business.

- Voluntary standards

There are a plethora of voluntary standards concerned with a wide range of things from business processes to technical specifications. The extent of their impact on the design process is equally wide ranging from the processes used during development to the availability of suitable components that can be adapted for use rather than designed from scratch.

- Industry codes of practice

Sometimes termed 'self regulation', this is where the organisations interested in an industry sector agree common ways of doing things in order to increase the acceptability of their products and services to their consumers. The interested organisations can include users as well as suppliers. These 'rules' do not have the force of law, but in some cases there are monitoring bodies that can impose punitive measures. Clearly, any development needs to be able to accommodate any relevant codes of practice.

- Consumer acceptability

Less formal, but more critical to unlocking the full potential of electronic commerce, the design needs to accommodate the needs and expectations of the potential consumer (e.g. MS DOS command prompts may not be the most acceptable user interface for interactive television).

Although these classifications do prove useful, it is worth noting at this early stage that there is a lot of interaction between the types of issue they represent. Examples of this interaction are:

- industry codes of practice may include the way in which particular voluntary standards are applied to the industry,
- regulation is often based on the requirement to conform to a number of specified 'voluntary' standards, or even the requirement to follow the code of practice managed by a specified body.

3. Which organisations discuss these issues?

Clearly, since these issues have such a large impact on what BT, or any developer, can do, there needs to be an awareness, not only of the current status of these constraints, but also of any planned changes. In addition, BT must be prepared to explore opportunities to influence the future environment by participating in and contributing towards the formulation of agreements that create these external constraints. As might be expected, the range of avenues that can be used to gain information on proposed changes to, and provide BT's input into the formulation of, the external environment is as wide as the range of issues themselves.

Voluntary standards are created in a number of ways and documented in a variety of forms. These forms can be grouped into two major classes.

- Formal standards

These are agreements documented by a formally recognised standards body. Such bodies have rigid procedures to reach agreements and the resulting standards are often sold as a revenue stream for the standards body. Formal bodies that reach agreements spanning more than one country (e.g. CEN (European) or ITU (global)) often reach their agreements through nationally oriented voting procedures and are used to ensure openness on international trade.

- Informal standards

Industry consortia and forums develop informal, or industry, standards. They are characterised by the fact that agreements are reached between the members (rather than countries) and they mostly address more direct practical issues of implementation. Indeed many such agreements are focused on recommending a combination of formal standards, or a way of implementing the options in a formal standard in order to meet a specific requirement.

Both of these types of standard can affect the range of devices that can be procured. Formal standards can also form the basis of regulations, and both formal and informal standards can be the basis of a code of practice. Clearly these types of organisation are a key avenue for BT influence. However, there are many organisations producing these documents and it is not always obvious which are the ones that should be attended. Formal standards have a status in their own right, but, if they are not used by industry, nor referenced by regulators, there is little value in influencing their content. On the other hand, an industry grouping that does not include the key players in the sector is unlikely to produce agreements that are used. Since these informal standards only have value if they are used, participation in poorly supported groups is likely to be

a waste of resources. Choosing where to apply BT's resources can be a difficult issue to resolve.

Regulations are often based on advice drawn from knowledgeable individuals and groups. Clearly, in the regulatory areas, where there is an interest, it is important to gain access to such groups. Equally, there is often a consultation process during the formulation of legislation, and responding to them is an important opportunity. One of the important pieces of information that can be gained from these is to understand which voluntary standards are being referenced, and hence in which groups to participate.

Consumer acceptability is very different and needs to be approached from an alternative angle. The most common approach is through awareness campaigns, possibly utilising trial results. These campaigns can be undertaken by individual organisations to promote their solutions, or by industry groups aimed at promoting a whole market sector. In some cases, as a single organisation can be involved in both informal standards generation and awareness programmes, the rationale for participation in an organisation may not be purely technical. However, these are primarily marketing issues and are not treated further here.

4. Electronic commerce issues

These observations are true for all business sectors. When we consider eCommerce against these criteria then there are a number of characteristics for this area that influence the choices that are made. One obvious characteristic is the speed with which the Internet-based element is developing. Both the technologies involved and the business models are changing rapidly. This means that any necessary industry agreements for the area of eCommerce must also be reached quickly and militates

against using formal standardisation processes. On the other hand some elements, such as those looking to influence legislation and regulation, may well need to use the slower, formal processes in order to gain an agreement with adequate status so that it can be referenced.

In addition, eCommerce introduces many areas of complexity as it is dealing with multinational transactions, and the services used to support such trading relationships. As a result, the services and products need to accommodate not only the legal framework associated with such trade, but also any regulatory regimes that may impose requirements on an element of the service (e.g. funds transfer).

The resulting mix of organisations is complex and not easily described in isolation. It is easier to take a few issues and consider the range of bodies that may be involved in each.

4.1 Electronic signatures

Rea [1] considers the technical issues associated with digital signatures and certification authorities. However, as digital signatures imply a specific technology, where work in this area needs to be technology independent the term 'electronic signatures' is used. The range of external activities is very wide including technology-specific and technology-independent aspects.

The major focus currently is the range of legislation and regulation that is forming at both national and European levels (see Fig 1). This activity is essential to the establishment of eCommerce. If eCommerce is to be effective then electronic signatures must have legal recognition, and this is at the core of this activity. There are

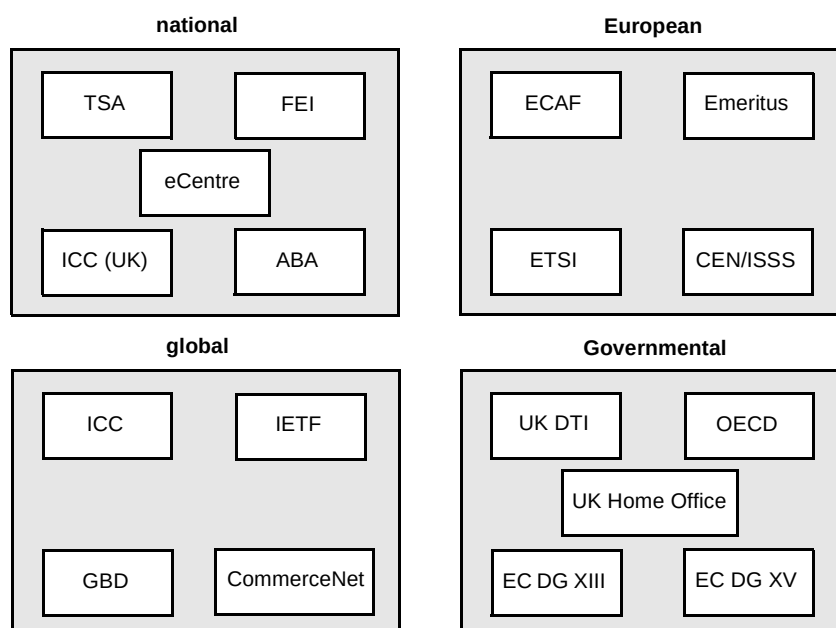


Fig 1 Electronic signature regulation and legislation.

obviously quality issues and much of the detail is associated with establishing minimum requirements on the service providers (like certification authorities) that support the digital signature infrastructure.

At the European level, the work is focusing around the drafting of a directive on electronic signatures [2] which was developed with the involvement of Directorates General 13 and 15 of the European Commission. BT has commented on this document throughout its development. The major issue of concern is around the regulatory framework that the directive aims to establish. The regulations focus on the requirements needed to assure the 'quality' of the systems supporting the electronic signature. There are debates regarding the extent to which the directive can be independent of technology, and the balance between formal regulation and industry-based self-regulation. In order to support the emphasis on self-regulation, BT is involved in a number of industry-led bodies. These bodies explore the avenues open to establish practical proposals for such a framework of self-regulation. There are a number of such initiatives, including bodies like the ICC (International Chamber of Commerce), ECAF (European Certification Authority Forum), TSA (Trust Service Association), or ETSI (European Telecommunications Standards Institute). Indeed the TSA is part of a group working on the Emeritus project that is part-funded under the European TEN Telecom (Trans-European Telecom-munications Networks) programme.

At the national level, there are different situations in each country. The major activities are beginning to address the national issues arising from the European directive and to assess the necessary changes to existing laws that require signatures to be in 'writing'. BT has been contributing to the discussions on what needs to be done in the UK, including providing comments on the current consultation paper [3] (that has been developed by DTI and the Home Office), and will take future opportunities to comment on further proposals. BT will take opportunities to comment directly and also to contribute to the submissions from trade associations, like FEI (Federation of the Electronics Industry) or eCentre [4], of which it is a member.

An associated issue is how will the certification authorities (CAs) make their certification policy and practice statements available to all who need them (potentially anyone who relies on the certificate). Clearly an electronic means must be found. One option is to include them as statements within the certificate, but this will make the certificate very large, probably too large for it to be held on a smart card. An alternative is to make this information available over the Internet, but users need to be sure that the information is managed correctly, and changes are correctly dated. BT has been working closely with the International Chamber of Commerce to establish a managed document repository. The intention is that documents such as certi-

fication policies and certificate practice statements can be lodged in the repository but the management of the documents is under the control of the ICC. In this way it is hoped that the limitation of smart card technology can be addressed by creating an appropriate external environment.

4.2 Contracting

While the legal recognition of an electronic signature is a key step forward, the ability to enter into a contract on-line will be a key enabler of electronic commerce. This issue is very complex, not least because binding contracts can be made today on the Internet; in only a few cases is the medium (i.e. paper) a legal requirement. The major difficulty arises because the nature of the Internet means that many of the contracts are likely to be created between parties in different countries, with different legal systems, and it is important to understand under which jurisdiction the contract is made.

International trade is not new and there are some recognised rules, but applying these can be less than clear. For example, is a contract established via e-mail treated according to postal rules (it exists at the point where the accepting message enters the mail system) or instantaneous rules (e.g. fax, it exists at the point of receipt of the message). Since the two ends could be in different countries, establishing acceptable conventions on how to apply such rules will be important to the future of eCommerce.

When dealing with this issue, there are different legal situations depending on whether you are considering a contract between businesses, or a contract involving consumers (see Fig 2).

Business-to-Business

Historically, international trade rules only considered business-to-business contracts. As such, the legal situation for electronic contracting is helped by the existing framework put in place by UNCITRAL (United Nations Commission on International Trade Law) and supported by the activities of the ICC. An example of this is the fact that the contract can state the legal jurisdiction under which it has been created. Such freedoms will greatly simplify the creation of an Internet friendly environment.

Consumer

The issue of consumer contracts is much more fraught. Such contracts are subject to a range of consumer protection legislation that has been created on a national basis working on the assumption that international consumer trade is negligible. This assumption is being challenged by the Internet and causing a number of problems. For example, it has been suggested that the jurisdiction should be that of

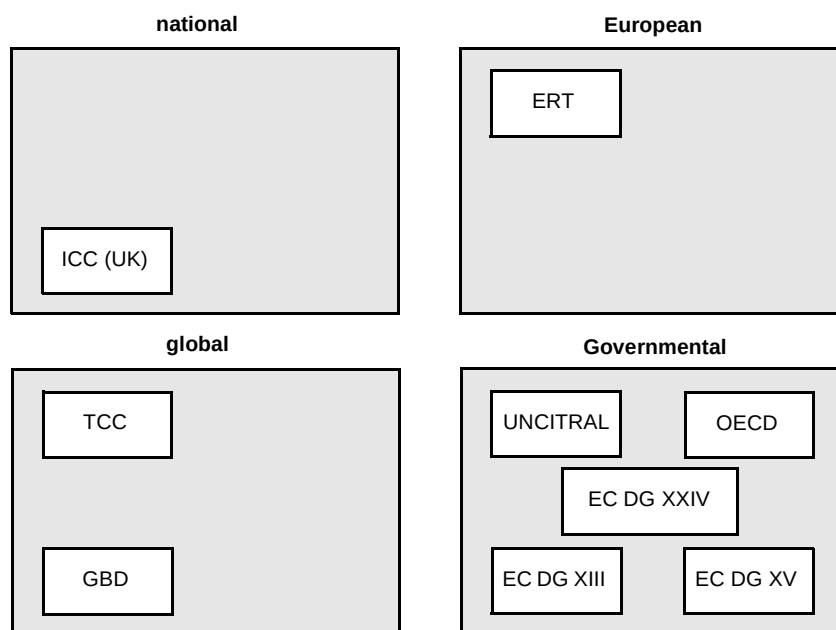


Fig 2 Contracting regulation and legislation.

the consumer and that contract clauses that attempt to state a different jurisdiction for the contract (as would be acceptable between businesses) are unfair trading terms and hence unenforceable. While this may be logical from a legal perspective, it would mean that any Web site that offers to sell goods will be subject to all legal jurisdictions. Such implications are not just restricted to contracts. If a Web site is classed as advertising, it would also become subject to all the advertising restrictions in the world.

Such uncertainties are being addressed at all levels including the European Round Table on Consumer Issues (ERT) and the Global Business Dialogue (GBD). However, the debates are likely to be highly political as well as technically difficult and may delay the large-scale take-up of Internet trading with consumers. Such a wide-ranging issue falls within a number of DGs within the European Commission and indications of the range of sensitivities can be found in the proposed European Directive on Legal Aspects of Electronic Commerce [5].

4.3 EDI futures

EDI is the longest established standardised technology in the electronic commerce arena. Over the years of its development, EDI has become a good fit for its users. However, the range of commercial transactions that it is now perceived as possible to address electronically is much larger and 'traditional EDI' is incapable of addressing these newer opportunities. As a result both traditional and newer bodies have started to consider how the most relevant bits of EDI can be exploited in these new areas. There are a number of different thrusts, but they all begin by separating the standardised message formats of EDI from the VANS (value added network services) currently used to deliver

EDI-based services to today's customers. Three examples of these activities are given below.

- SimplEDI — a new approach to EDI messages

Instead of repeating all the information in every message, 'static' information (i.e. the information that does not change, like company names and addresses) is held in a common database and referenced as required. This simplifies the messages and reduces implementation costs.

- EDI over Internet

This considers the appropriate ways for conveying EDI-format messages over Internet network technologies. This provides a way forward for companies with large investments in current EDI.

- XML/EDI

This format is looking towards the use of EDI to convey the information gathered by Web forms to remote database systems, and vice versa.

As might be expected, there are high levels of activity in 'traditional' bodies like UKCEDIS (UK Confederation for EDI Standards) and CEFAC (Centre for Facilitation of Procedures and Practices for Administration, Commerce and Transport). However, it would be misleading to suggest the newer bodies like the IETF (Internet Engineering Task Force) and CommerceNet are undertaking the more innovative activities (see Fig 3). In this area there are a number of possible ways that the industry may go, and they are not necessarily mutually exclusive.

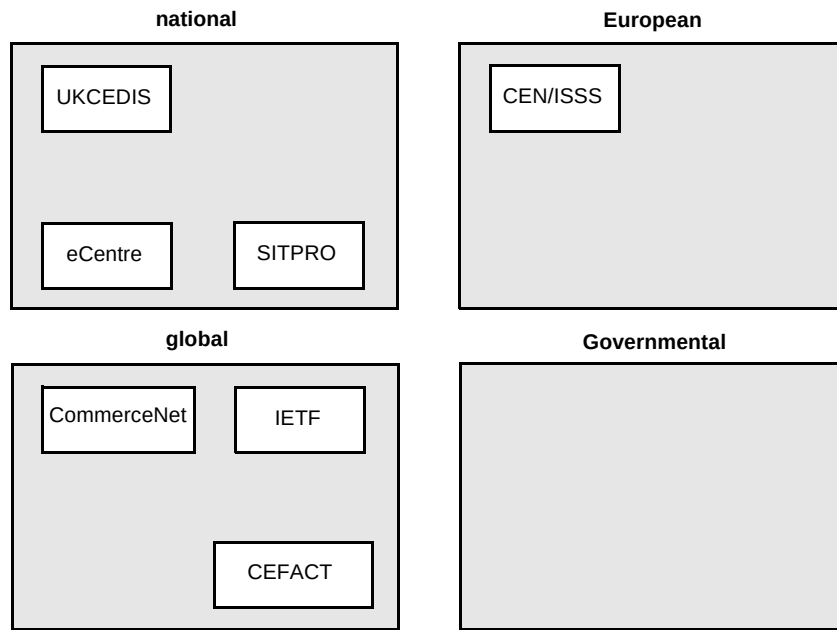


Fig 3 EDI futures regulation and legislation.

5. Conclusions — woods, trees and chocolate elephants

As can be seen from this paper, there are many issues related to electronic commerce that are affected by decisions in external organisations. It is not unexpected that the breadth of issues (from technical to legal and national to global) means that the debates are undertaken in a range of bodies. Indeed, the scope of the activity is so large the only way of approaching (like eating a chocolate elephant) is by taking one bite at a time.

On the other hand, the fast moving nature of this area means that many of these activities are progressing in parallel. As a result, care must be taken to ensure that when the individual agreements, perfectly crafted trees though they may be, are brought back together the overall effect is a wood which creates the environment within which electronic commerce can flourish.

References

- 1 Rea T: 'High value certification — trust services for complex eCommerce transactions', BT Technol J, 17, No 3, pp 50—56 (July 1999).

- 2 European Commission: 'Proposal for a European Parliament and Council Directive on a common framework for electronic signatures; Communication from the Commission to the European Parliament, the Council, the Economic and Social Committee and the Committee of the Regions; COM (1998) 297 final', (13 May 1998).
- 3 DTI/Home Office: 'Public Consultation Paper on Licensing of Trusted Third Parties for the provision of Encryption Services', (1999).
- 4 <http://www.ecentre.org.uk/>
- 5 European Commission: 'Proposal for a European Parliament and Council directive on certain legal aspects of electronic commerce in the internal market; Com (1998) 586 Final', (1998).



Paul Jenkins has been involved in data communications for over 15 years. He has held a number of international posts within the standards environment including roles within ISO and ITU. More recently he has been involved in establishing BT's strategic approach to emerging technologies, beginning with multimedia and then focusing in on electronic commerce.

He is currently responsible for BT's technical strategy for electronic commerce and determining the voluntary eCommerce standards bodies in which BT will be involved.